EXPERIMENT - 6

Aim:-

Create a stored procedure transfer_employee(emp_id, new_dept_id) with validation and error handling. Implement triggers for salary validation and audit logging. Test edge cases.

Source Code:-

Audit Table

```
CREATE TABLE Employee_Audit (
   audit_id INT AUTO_INCREMENT PRIMARY KEY,
   emp_id INT,
   old_dept_id INT,
   new_dept_id INT,
   old_salary DECIMAL(10,2),
   new_salary DECIMAL(10,2),
   action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Salary Validation Trigger

```
CREATE TRIGGER validate_salary
BEFORE INSERT ON Employee
FOR EACH ROW
IF NEW.salary <= 0 THEN
   SIGNAL SQLSTATE '45000'
   SET MESSAGE_TEXT = 'Salary must be greater than 0';
END IF;
```

Salary Validation for Update

```
CREATE TRIGGER validate_salary_update
BEFORE UPDATE ON Employee
FOR EACH ROW
IF NEW.salary <= 0 THEN
```

```
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Salary must be greater than 0';
END IF;
```

Audit Logging Trigger

```
CREATE TRIGGER employee_audit
AFTER UPDATE ON Employee
FOR EACH ROW
IF OLD.dept_id <> NEW.dept_id
   OR OLD.salary <> NEW.salary THEN

    INSERT INTO Employee_Audit
    (emp_id, old_dept_id, new_dept_id, old_salary, new_salary)
    VALUES
    (OLD.emp_id, OLD.dept_id, NEW.dept_id,
     OLD.salary, NEW.salary);
END IF;
```

Stored Procedure - Transfer Employee

```
CREATE PROCEDURE transfer_employee(
    IN p_emp_id INT,
    IN p_new_dept_id INT
)
BEGIN
    DECLARE employee_exists INT DEFAULT 0;
    DECLARE department_exists INT DEFAULT 0;

    SELECT COUNT(*)
    INTO employee_exists
    FROM Employee
    WHERE emp_id = p_emp_id;

    SELECT COUNT(*)
    INTO department_exists
    FROM Department
    WHERE dept_id = p_new_dept_id;

    IF employee_exists = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee does not exist';
```

```
    ELSEIF department_exists = 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Department does not exist';

    ELSE
        UPDATE Employee
        SET dept_id = p_new_dept_id
        WHERE emp_id = p_emp_id;
    END IF;
END
```

Test Stored Procedure

```
CALL transfer_employee(1, 5);
```

Checking Employee

```
SELECT emp_id, emp_name, dept_id
FROM Employee
WHERE emp_id = 1;
```

Test Audit Log

```
SELECT *
FROM Employee_Audit;
```

Test Salary Validation

```
UPDATE Employee
SET salary = 5000
WHERE emp_id = 1;

SELECT *
FROM Employee
WHERE emp_id = 1;
```

Test Invalid Employee

```
CALL transfer_employee(999, 2);
```

Test Invalid Department

```
CALL transfer_employee(101, 99);
```

Output:-

TRANSFER EMPLOYEE

fieldCount	affectedRows	insertId	info	serverStatus	warningStatus	changedRows
0	1	0		2	0	0

EMPLOYEE AFTER TRANSFER

emp_id	emp_name	dept_id
1	Aarav Sharma	5

AUDIT LOG

audit_id	emp_id	old_dept_id	new_dept_id	old_salary	new_salary	action_time
1	1	1	3	65000.00	65000.00	2026-09-27T08:47:33.000Z
2	1	3	2	65000.00	65000.00	2026-09-27T08:48:09.000Z
3	1	2	5	65000.00	65000.00	2026-09-27T08:50:11.000Z

SALARY VALIDATION

fieldCount	affectedRows	insertId	info	serverStatus	warningStatus	changedRows
0	1	0	Rows matched: 1 Changed: 0 Warnings: 0	2	0	0

emp_id	emp_name	gender	salary	hire_date	dept_id	project_id	manager_id
1	Aarav Sharma	Male	5000.00	2022-01-15T00:00:00.000Z	5	101	null

TEST INVALID EMPLOYEE

```
Error: Employee does not exist
```

TEST INVALID DEPARTMENT

```
Error: Employee does not exist
```